

Adaptation Needs in Robotic Systems: Behavior Trees and Beyond (Questionnaire)

You are invited to participate in a research questionnaire about adaptation needs in robotic systems that use Behavior Trees.

The purpose of this study is to collect opinions from **researchers** and **practitioners** with experience in robotics, behavior trees, autonomous systems, or related areas. In particular, we are interested in:

1. how relevant different adaptation needs are for robotic missions encoded using Behavior Trees; and
2. how suitable different Behavior Tree adaptation approaches are for addressing those needs.

The questionnaire should take approximately **15–20 minutes** to complete. Participation is voluntary. You may stop at any time before submitting the questionnaire.

Data processing

Any personal information will be removed during analysis and will not appear in any publications or presentations. Only the research team will have access to contact data, which will be used solely for study-related communication.

Ethical aspects

No personal, sensitive, or potentially harmful information is collected. Participation is voluntary, and no risks or direct benefits are expected. Your responses are anonymous and will be used solely for academic research purposes.

Consent

Your participation is voluntary and based on informed consent. You may withdraw at any time by closing the survey before submission. Upon request, all data associated with you will be deleted and excluded from any analysis or publication.

Contact Information

If you have any questions or concerns about the study, please contact us at:

- Mehran Rostamnia: mehran.rostamnia@gssi.it (**Main Contact**)
- Ricardo Caldas: ricardo.caldas@gssi.it
- Gianluca Filippone: gianluca.filippone@gssi.it
- Patrizio Pelliccione: patrizio.pelliccione@gssi.it

* Indicates required question

1. Informed Consent *

Please confirm all the following before continuing.

Tick all that apply.

- ☐ I have read the study information above.
- ☐ I understand that participation is voluntary.
- ☐ I understand that I may stop before submitting the questionnaire.
- ☐ I consent to participate in this study.

Section 1: Background

This section asks about your background. Please answer based on your own experience. "Experience" may include research, industrial work, coursework, software development, simulation, field experiments, system integration, or personal projects.

2. How would you rate your experience in Robotics? *

1 = No experience / Novice: little or no prior experience (< 1 year)

3 = Intermediate: some practical or theoretical familiarity (1 -- 3 years)

5 = Expert: substantial experience, typically several years of study, research, or practice (> 3 years)

Mark only one oval.

	1	2	3	4	5	
Nov	☐	☐	☐	☐	☐	Expert

3. How would you rate your experience in Behavior Trees? *

1 = No experience / Novice: little or no prior experience (< 1 year)

3 = Intermediate: some practical or theoretical familiarity (1 -- 3 years)

5 = Expert: substantial experience, typically several years of study, research, or practice (> 3 years)

Mark only one oval.

	1	2	3	4	5	
Nov	☐	☐	☐	☐	☐	Expert

4. How does your work relates to Behavior Trees? *

Mark only one oval.

- ☐ I have used Behavior Trees to implement robotic behavior.
- ☐ I have developed tools, methods, or algorithms that use Behavior Trees.
- ☐ I have studied or researched Behavior Trees as a formalism for robotic behavior.
- ☐ I have worked on projects that use Behavior Trees, but I have not modified or implemented them myself.
- ☐ I am familiar with Behavior Trees, but I have not used them directly.
- ☐ I am not familiar with Behavior Trees.
- ☐ Other: _____

5. In which type of organization have you worked with robotics, Behavior Trees, or autonomous systems? *

Tick all that apply.

- ☐ Academia
- ☐ Industry
- ☐ Government or public-sector organization
- ☐ Research Institute
- ☐ Open-source or community project
- ☐ Independent / personal projects
- ☐ I have not worked with them in an organizational setting
- ☐ Other: _____

6. What robotics domains have you worked with? *

Tick all that apply.

- ☐ Service robotics
- ☐ Marine robotics
- ☐ Space robotics
- ☐ Field robotics
- ☐ Healthcare robotics (e.g., prosthetics, surgery)
- ☐ Industrial automation
- ☐ Construction
- ☐ I have not worked in a specific robotics domain
- ☐ Other: _____

7. What is your main area of expertise or role?

Examples: robotics researcher, software engineer, autonomy engineer, PhD student, system integrator, robotics architect, safety engineer.

Section: Adaptation Needs in Robotics

In this questionnaire, an **adaptation need** means a situation in which a robotic system may need to update, repair, specialize, replace, or otherwise change its current behavior, knowledge, configuration, or mission interpretation in order to maintain or improve mission success, safety, reliability, robustness, or efficiency.

In the following questions, please evaluate **how relevant each adaptation need is to robotic systems and how effectively those can be addressed using Behavior Trees.**

- **Not relevant:** unlikely to require adaptation of the robotic systems.
- **Slightly relevant:** may require adaptation in limited cases.
- **Moderately relevant:** can require adaptation in some realistic cases.
- **Very relevant:** often requires adaptation in realistic robotic systems.
- **Extremely relevant:** central or critical reason for adaptation.
- **Insufficient expertise:** you do not feel able to judge this item.

8. How relevant are the following **knowledge-related adaptation needs to robotic systems?** *

Knowledge-related adaptation needs consist of knowledge abstraction, incompleteness, inaccuracy, and drift. Below, we describe each of them.

- **Knowledge abstraction** — the available knowledge is too abstract, coarse, or simplified for the robot's current task or execution context.
- **Knowledge incompleteness** — relevant information about the task, robot, objects, environment, or constraints is missing.
- **Knowledge inaccuracy** — the robot's internal information is incorrect, inconsistent, or based on wrong assumptions.
- **Knowledge drift** — information that was previously valid becomes outdated as the robot, task, or environment changes over time.

Mark only one oval per row.

	Not relevant	Slightly relevant	Moderately relevant	Very relevant	Extremely relevant	Insufficient expertise
Knowledge-related needs	☐	☐	☐	☐	☐	☐

9. How relevant are the following **perception-** and **actuation-related adaptation needs** to robotic systems? *

Perception-related adaptation needs consist of perception multiplicity, inaccuracy, and unavailability. **Actuation-related adaptation needs** consist of actuation nondeterminism and unavailability. Below, we describe each of these adaptation needs.

- **Perception multiplicity** – the robot receives multiple possible interpretations of the same situation, object, event, or state.
- **Perception inaccuracy** – sensor readings, detections, classifications, or state estimates are noisy, uncertain, or incorrect.
- **Perception unavailability** – required perceptual information is unavailable because of occlusion, sensor failure, limited coverage, or communication loss.
- **Actuation nondeterminism** – actions may produce variable or uncertain outcomes even when the same Behavior Tree node is executed.
- **Actuation unavailability** – an actuator, skill, controller, or motion capability becomes degraded, unavailable, or unsafe to use.

Mark only one oval per row.

	Not relevant	Slightly relevant	Moderately relevant	Very relevant	Extremely relevant	Insufficient expertise
Perception- and actuation- related needs	☐	☐	☐	☐	☐	☐

10. How relevant are the following **system-related adaptation needs** to **robotic systems**? *

System-related adaptation needs consist of component failure, changing resources, new resources, and communication impairment. Below, we briefly depict each corresponding adaptation needs.

- **Component failure** – a hardware or software component required by the Behavior Tree fails or becomes unreliable.
- **Changing resources** – available resources such as battery, time, computation, storage, bandwidth, or payload capacity change during execution.
- **New resources** – the robot gains access to new components, services, tools, skills, or external systems.
- **Communication impairment** – communication between robots, modules, operators, cloud services, or external systems becomes delayed, intermittent, or unavailable.

Mark only one oval per row.

	Not relevant	Slightly relevant	Moderately relevant	Very relevant	Extremely relevant	Insufficient expertise
system-related needs	☐	☐	☐	☐	☐	☐

11. How relevant are the following **mission-related adaptation needs** to **robotic systems**? *

Mission-related adaptation needs include uncertain mission specification, mission changes during execution, outdated mission, uncertain quality goals, and uncertain relationships. Each of these adaptation needs is briefly described below.

- **Uncertain mission specification** – the mission objective, success condition, or required behavior is incomplete, ambiguous, or underspecified.
- **Mission changes during execution** – the mission goal, task priority, operator instruction, or operational constraint changes while the robot is executing.
- **Outdated mission** – the Behavior Tree was designed for assumptions that no longer hold.
- **Uncertain quality goals** – tradeoffs among safety, time, energy, robustness, accuracy, or user preferences are unclear or change over time.
- **Uncertain relationships among goals** – dependencies, conflicts, or ordering constraints among subgoals are unclear or change during execution.

Mark only one oval per row.

	Not relevant	Slightly relevant	Moderately relevant	Very relevant	Extremely relevant	Insufficient expertise
mission-related needs	☐	☐	☐	☐	☐	☐

12. How relevant are the following **environment-related adaptation needs** to robotic systems? *

Environment-related adaptation needs consist of execution context changes, non-human agent coexistence, and human agents coexistence. Below, we briefly explain each of these adaptation needs.

- **Execution context changes** – relevant environmental conditions change, such as layout, terrain, lighting, weather, obstacles, object locations, or accessibility.
- **Non-human agents coexistence** – the robot must account for other robots, autonomous systems, or agents whose behavior affects execution.
- **Human agents coexistence** – the robot must account for humans who may collaborate with, supervise, obstruct, depend on, or be affected by the robot.

Mark only one oval per row.

	Not relevant	Slightly relevant	Moderately relevant	Very relevant	Extremely relevant	Insufficient expertise
Environment-related needs	☐	☐	☐	☐	☐	☐

13. Please briefly explain any concerns or comments regarding the adaptation needs listed above, particularly if you consider any of them important, unimportant, relevant, irrelevant, incomplete, or unclear.

14. In your opinion, how should each family of adaptation needs primarily be addressed using BTs? *

You can find the definitions of the adaptation needs on [this page](#).

Mark only one oval per row.

	Using classical BTs is enough	Mainly by extending the BT formalism	Mainly by updating the BT, without changing the formalism	Mainly by adapting external modules/models	By both adapting BTs formalism and external modules	None of the above (explain)
Knowledge-related needs	☐	☐	☐	☐	☐	☐
Perception- and actuation-related needs	☐	☐	☐	☐	☐	☐
System resource and architecture needs	☐	☐	☐	☐	☐	☐
Mission- and goal-related needs	☐	☐	☐	☐	☐	☐
Environment- and agent-related needs	☐	☐	☐	☐	☐	☐

15. Please briefly explain your answer, if you wish.

Section: Approaches and Techniques to enhance BTs for Adaptation in robotic systems

This section asks about four approaches for adapting BTs in robotic systems. Please judge each approach in terms of its **suitability** for addressing different families of adaptation needs in robotic systems.

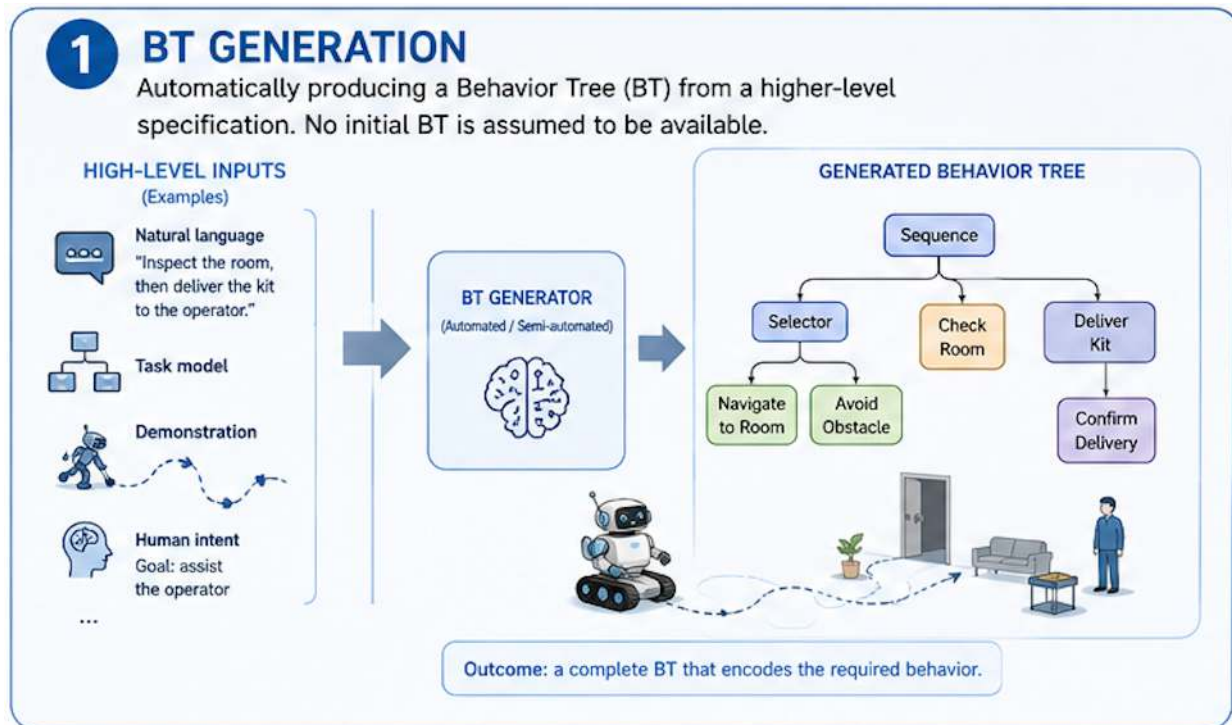
Here, **suitability** means your assessment of whether the approach is **technically appropriate** and **practically useful** for addressing the need in robotic systems using BTs. Please consider the approach in general, not only your personal experience with it.

Use the scale as follows:

- **Not suitable:** the approach is generally inappropriate for this need.
- **Partially suitable:** the approach may help in limited cases.
- **Completely suitable:** the approach is completely appropriate for this need.
- **Insufficient expertise:** you do not feel able to judge this item.

Definition: Behavior Tree Evolution Techniques

BT Generation refers to automated or semi-automated processes that produce a BT from a higher-level specification, such as human intent, task models, natural-language instructions, demonstrations, plans, or other informal descriptions. In BT generation, **no initial BT** is taken as the behavior to be modified, although existing BTs, BT fragments, templates, libraries, or design patterns may be used as prior knowledge to guide the generation process.



16. How **suitable** is **BT Generation** for addressing each family of adaptation needs in * robotic systems?

You can find the definitions of the adaptation needs on [this page](#).

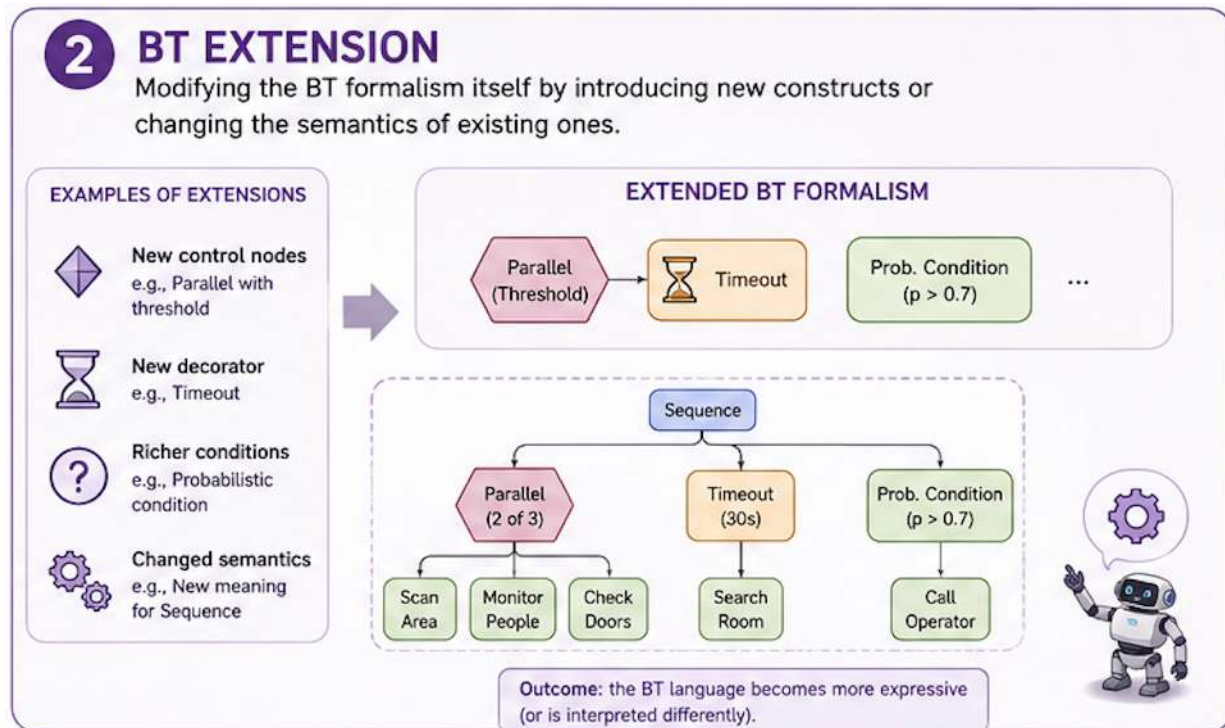
Mark only one oval per row.

	Not suitable	Partially suitable	Completely suitable	Insufficient expertise
Knowledge-related needs	☐	☐	☐	☐
Perception- and actuation-related needs	☐	☐	☐	☐
System resource and architecture needs	☐	☐	☐	☐
Mission- and goal-related needs	☐	☐	☐	☐
Environment- and agent-related needs	☐	☐	☐	☐

17. Please briefly explain any case where you consider **BT Generation** to be *Not suitable* or *Partially suitable* for addressing an adaptation need.

Definition: Behavior Tree Extension Techniques

BT Extension refers to processes that modify the BT formalism itself. This may involve introducing new syntactic constructs into the BT grammar, changing the semantics of existing constructs, or both. Unlike generation, evolution, or refinement, BT extension changes what can be expressed within the BT language or how BT nodes are interpreted.



18. How **suitable** is **BT Extension** for addressing each family of adaptation needs in robotic systems? *

You can find the definitions of the adaptation needs on [this page](#).

Mark only one oval per row.

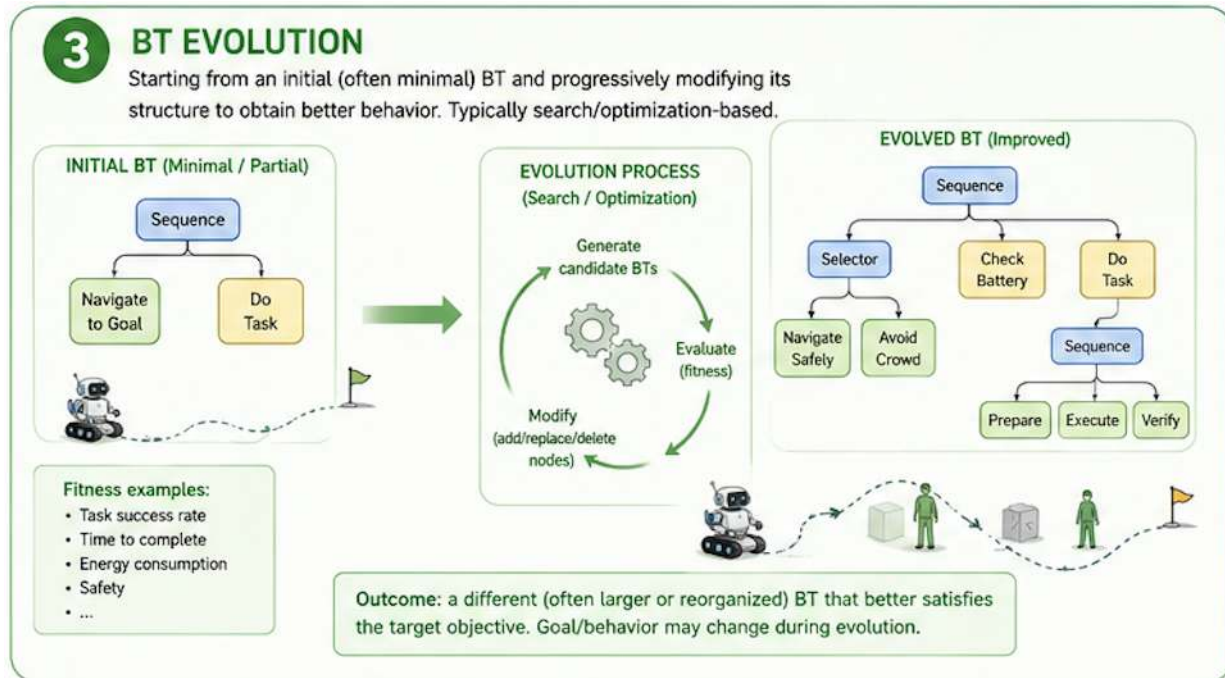
	Not suitable	Partially suitable	Completely suitable	Insufficient expertise
Knowledge-related needs	☐	☐	☐	☐
Perception- and actuation-related needs	☐	☐	☐	☐
System resource and architecture needs	☐	☐	☐	☐
Mission- and goal-related needs	☐	☐	☐	☐
Environment- and agent-related needs	☐	☐	☐	☐

19. Please briefly explain any case where you consider **BT Extension** to be *Not suitable* or *Partially suitable* for addressing an adaptation need.

Definition: Behavior Tree Evolution Techniques

BT Evolution refers to processes that start from an initial, often minimal or partial, Behavior Tree and progressively modify its structure to obtain a tree that encodes more complex or improved behavior.

These modifications may include adding, replacing, deleting, or reorganizing nodes. Evolution is commonly associated with search-based or optimization-based techniques, such as genetic programming, where candidate **Behavior Trees are iteratively improved according to a fitness criterion**.



20. How suitable is **BT Evolution** for addressing each family of adaptation needs in robotic systems? *

You can find the definitions of the adaptation needs on [this page](#).

Mark only one oval per row.

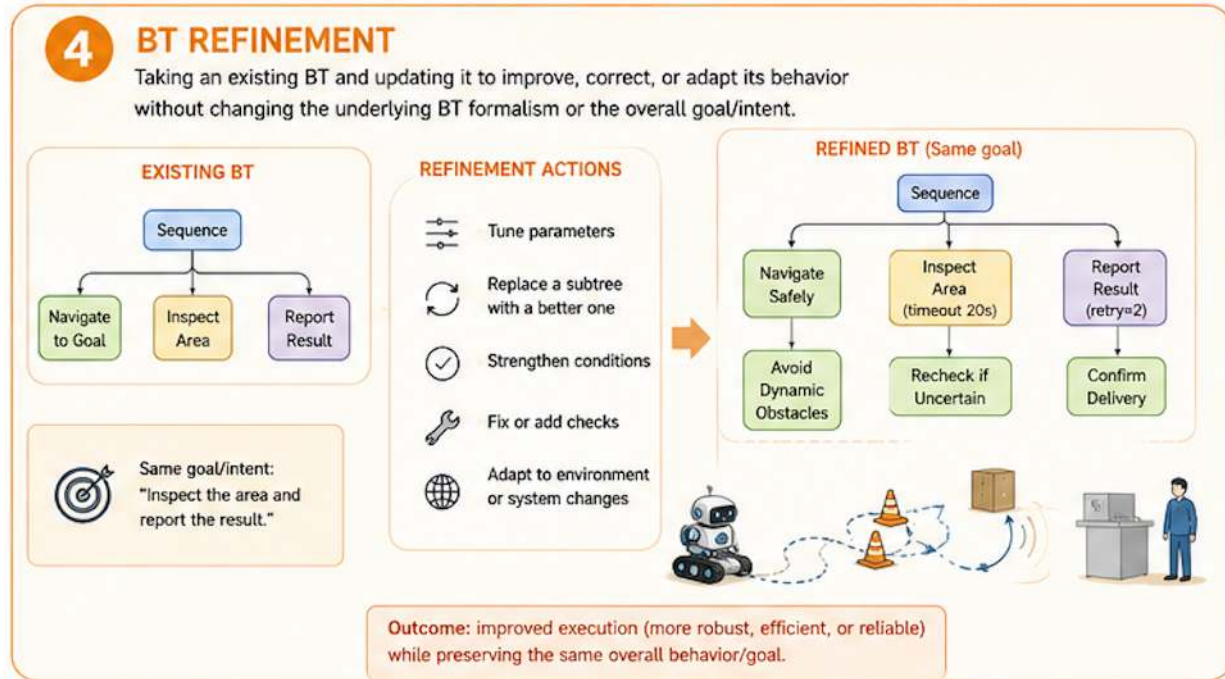
	Not suitable	Partially suitable	Completely suitable	Insufficient expertise
Knowledge- related needs	☐	☐	☐	☐
Perception- and actuation- related needs	☐	☐	☐	☐
System resource and architecture needs	☐	☐	☐	☐
Mission- and goal-related needs	☐	☐	☐	☐
Environment- and agent- related needs	☐	☐	☐	☐

21. Please briefly explain any case where you consider **BT Evolution** to be *Not suitable* or *Partially suitable* for addressing an adaptation need.

Definition: Behavior Tree Refinement

BT Refinement refers to processes that take an existing, meaningful Behavior Tree as input and update it to improve, correct, or adapt its behavior without changing the underlying Behavior Tree formalism.

Refinement may modify specific nodes, subtrees, parameters, guards, preconditions, postconditions, recovery actions, or execution conditions. In this questionnaire, BT Refinement is mainly about **adjusting an existing behavior while preserving the same overall goal or intended task**.



22. How suitable is **BT Refinement** for addressing each family of adaptation needs in robotic systems?

You can find the definitions of the adaptation needs on [this page](#).

Mark only one oval per row.

	Not suitable	Partially suitable	Completely suitable	Insufficient expertise
Knowledge-related needs	☐	☐	☐	☐
Perception- and actuation-related needs	☐	☐	☐	☐
System resource and architecture needs	☐	☐	☐	☐
Mission- and goal-related needs	☐	☐	☐	☐
Environment- and agent-related needs	☐	☐	☐	☐

23. Please briefly explain any case where you consider **BT Refinement** to be *Not suitable* or *Partially suitable* for addressing an adaptation need.

Section 5: Additional Comments

Optional and open questions in case you have complimentary information to share.

24. Are there any adaptation needs that you think are missing from this questionnaire?

25. Are there any comments, qualifications, or concerns you would like to add about Behavior Tree adaptation in robotic systems?

Section 6: Optional Contact Information

Providing contact information is optional. Your questionnaire responses are useful even if you prefer not to provide contact details.

If you provide an email address, we will use it only for study-related communication, such as sharing study results or asking whether you would be willing to participate in follow-up discussion.

26. To receive the results of the study, please enter your email address.

Optional. We will use this only to contact you about this study or future research activities.

27. May we contact you for follow-up about your answers?

Mark only one oval.

- ☐ Yes, you may contact me about my questionnaire responses.
- ☐ Yes, you may contact me only to share study results.
- ☐ No, please do not contact me.

Thanks a lot for your time and contribution.

Please click **Submit** to finish the survey.

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